1 Dynamic viscosity, μ, measures a fluid's internal resistance to gradual deformation by shear or tensile stress. The SI unit for dynamic viscosity is newton second per square meter (N s m^{-2}).

Given that

$$\mu = \frac{vL\rho}{R}$$

where v is the velocity of the fluid, L the characteristic length and ρ is the density of the fluid.

What is the unit of R, expressed in SI units?

A kg N^{-1} m^{-1} **B** kg N^{-1} m^{-3} s^{-2} **C** m^{-2} s^{-2} **D** dimensionless